

Personality traits, text signals, and cross-framework correlations: a reference for LLM research

Psychological personality traits leave detectable but modest fingerprints in written text. The meta-analytic evidence converges on a clear picture: individual LIWC categories correlate with Big Five traits at $r \approx .05-.15$, [\(Vrije Universiteit Amsterd...\)](#) [\(Matthewbjane\)](#) while machine learning over open-vocabulary features reaches $r \approx .30-.41$ [\(ResearchGate\)](#) — approaching the practical ceiling set by self-report reliability. [\(UC Irvine Language Science\)](#) For LLM finetuning, the most actionable insight is that personality traits cluster into predictable higher-order structures, cross-framework mappings are well-documented, and specific linguistic markers — though individually weak — form robust aggregate signatures when combined. This document synthesizes meta-analyses and large-sample studies across all major personality frameworks to serve as a hypothesis-generation reference for finetuning experiments and correlation checks.

1. Big Five (OCEAN): the foundational scaffold

Inter-trait correlations

The Big Five traits are **not orthogonal**. Meta-analysis (van der Linden et al., 2010; K=212, N=144,117) reveals consistent intercorrelations [\(ResearchGate\)](#) [\(ScienceDirect\)](#) at $r \approx .15-.30$, varying substantially by instrument. The NEO-FFI produces near-zero correlations between some pairs while the BFI and IPIP show larger intercorrelations.

Trait pair	Direction	Typical r
Agreeableness–Conscientiousness	Positive	.15–.30
Extraversion–Openness	Positive	.15–.30
Agreeableness–Neuroticism	Negative	–.15 to –.30
Conscientiousness–Neuroticism	Negative	–.20 to –.35
Extraversion–Neuroticism	Negative	–.10 to –.25
Agreeableness–Extraversion	Positive	.10–.20

These intercorrelations produce two well-documented higher-order factors. [Frontiers](#) **Stability (Alpha)** captures the shared variance of Agreeableness, Conscientiousness, and reversed Neuroticism [ScienceDirect](#) [Umn](#) — reflecting impulse control, social harmony, and emotional regulation, [isonderhouden](#) linked to serotonergic functioning. [ScienceDirect](#) **Plasticity (Beta)** captures the shared variance of Extraversion and Openness [ScienceDirect](#) [Umn](#) — reflecting exploration, engagement, and novelty-seeking, [isonderhouden](#) linked to dopaminergic functioning [ScienceDirect](#) (DeYoung, Peterson, & Higgins, 2002). These metatraits correlate at approximately $r \approx .45$, [isonderhouden](#) raising debate about a General Factor of Personality (GFP) at the apex. The GFP explains roughly 45% of Big Five variance in meta-analysis [ResearchGate](#) but is highly contested — critics argue it reflects socially desirable responding rather than genuine behavioral covariance, [ScienceDirect](#) and it vanishes when evaluative item content is removed (Bäckström et al., 2009) [Sage Journals](#) or when HEXACO measures are used (De Vries, 2011). [ScienceDirect](#)

At the intermediate level, DeYoung, Quilty, & Peterson (2007) identified **10 aspects** — two per Big Five domain — with distinct genetic substrates: [Jordan B. Peterson](#) Enthusiasm/Assertiveness (E), Compassion/Politeness (A), [Wiley Online Library](#) Industriousness/Orderliness (C), Volatility/Withdrawal (N), and Intellect/Openness (O). Cross-domain facet correlations at this level largely explain the higher-order structure and are critical for understanding why certain trait combinations co-occur in text.

Non-WEIRD samples show **stronger** Big Five intercorrelations [Sage Journals](#) (Gurven et al., 2013), and only three factors replicate consistently across all lexical studies worldwide (De Raad et al., [Wikipedia](#) 2010) [Sage Journals](#) — suggesting the five-factor structure may be partially culture-bound.

LIWC and linguistic markers

The definitive meta-analysis is **Koutsoumpis et al. (2022, Psychological Bulletin)**: 31 samples, $N=85,724$. Self-reported personality $\times$ 52 LIWC categories explains **5.1%** of personality variance, with individual category correlations at $|\rho| = .08-.14$. Observer-reported personality $\times$ LIWC is dramatically stronger, explaining **38.5%** of variance at $|\rho| = .18-.39$. [ResearchGate](#) Only **20 of 260** possible Big Five $\times$ LIWC combinations correlated with both self- and observer-reports — these represent the genuine "kernel of truth" in text-based personality assessment. [Vrije Universiteit Amsterd...](#)

Yarkoni (2010) analyzed 694 bloggers averaging 115,000 words each:

[Positivepsychologynews](#) [nih](#) 44% of Big Five $\times$ LIWC correlations were significant, with mean $|r| = .14$ and maximum single $r = .23$. [PubMed Central](#) **Schwartz et al. (2013)** used 75,000 Facebook users [PubMed](#) [PLOS](#) and showed open-vocabulary prediction ($r = .31-.41$) substantially outperforms LIWC ($r = .21-.29$). [UC Irvine Language Science](#) [ResearchGate](#)

Trait-by-trait linguistic profile:

Openness to Experience produces the richest linguistic signature: longer words (UCSC) (6+ letters, $r \approx .08-.15$), more articles and prepositions (markers of formal/complex prose), more tentative/hedging language ("perhaps," "maybe," UCSC $r \approx .08-.14$), more insight and cognitive process words, greater vocabulary diversity, and content about art, philosophy, literature, and existential themes. Low Openness manifests as concrete, practical language about sports, daily routines, and straightforward topics.

Neuroticism shows the single most robust personality–language association: more negative emotion words ($r \approx .10-.17$), more first-person singular pronouns ("I," "me," "my," UCSC $r \approx .08-.14$), Peggykern more anxiety and anger words, fewer positive emotion words, and content about personal struggles, health complaints, and relationship problems. Low Neuroticism (Emotional Stability) manifests as active lifestyle references, exercise, and sports content. (Positive Psychology Center)

Extraversion correlates with more social process words (Peggykern) (Columbia University) ($r \approx .07-.13$), positive emotion words ($r \approx .08-.15$), (Peggykern) simpler vocabulary, more informal language, exclamation marks, and intensifiers ("soooo"). Content centers on social activities, parties, friends, and entertainment. Low Extraversion (introversion) manifests as content about computers, internet culture, anime, reading, and solitary intellectual pursuits.

Agreeableness manifests as fewer swear words ($r \approx .08-.12$), (Peggykern) fewer anger words, more positive emotion language, (UCSC) more first-person plural pronouns ("we," "us"), and content about family, gratitude, faith, and community. Low Agreeableness shows profanity, antagonistic/competitive language, and blunt statements.

Conscientiousness shows the weakest linguistic markers overall: more achievement and work-related words, fewer swear words and negations, (UCSC) more future-tense verbs, and content about productivity, planning, and accomplishments. Low Conscientiousness shows boredom, procrastination references, and aimless content.

Text observability implications of higher-order structure

The Big Two metatraits translate directly into textual archetypes useful for LLM persona design. **High Stability** writers produce restrained, prosocial, organized, emotionally regulated text (PubMed Central) — less negative affect, more agreeable language, more achievement-oriented structure. **High Plasticity** writers produce exploratory, varied, creative, socially engaged text — more complex vocabulary, abstract content, diverse topics, enthusiastic social references. This two-dimensional simplification may be more tractable for LLM finetuning than five independent dimensions.

Effect sizes and replication status

Robust, well-replicated findings: Neuroticism → negative emotion words and first-person singular pronouns; Extraversion → positive emotion and social words; Agreeableness → fewer swear/anger words; (Peggykern) Openness → complex/long words and insight/cognitive words. These replicate across Pennebaker & King (1999), Mehl et al. (2006), Hirsh & Peterson (2009), Yarkoni (2010), (ScienceDirect) Schwartz et al. (2013), and Koutsoumpis et al. (2022).

Replication concerns: Martínez-Huertas et al. (2022) demonstrated a "failed cross-validation" of LIWC–personality relationships, (MDPI) and Berggren et al. (2024) showed that ML models trained on essays degrade when applied to Reddit. (Diva-portal) Effect size inflation in small samples is documented: Hirsh & Peterson (2009; N=94) reported mean significant $r = .23$, while Yarkoni's larger N=694 study found $r = .14$ — the expected power-inflation tradeoff. (PubMed Central) **Conscientiousness markers are the least reliable and most likely to fail replication.**

2. Dark Triad and Tetrad: low agreeableness as the common core

Internal structure and intercorrelations

The Dark Triad — narcissism, Machiavellianism, and psychopathy — shares a core of **callous-manipulation** that manifests as consistently positive intercorrelations. Vize et al. (2018; k=156, N=52,988) provides the most precise estimates: [PubMed](#) [ResearchGate](#) Machiavellianism–Psychopathy $\rho = .52$, Narcissism–Psychopathy $\rho = .38$, Machiavellianism–Narcissism $\rho = .35$. [ScienceDirect](#) The Machiavellianism–Psychopathy overlap is so large that Miller et al. (2017) found they share approximately **80%** of variance, raising serious questions about whether current Machiavellianism measures capture anything distinct from secondary psychopathy. [Frontiers](#)

Moshagen, Hilbig, & Zettler (2018) formalized this overlap as the **Dark Factor of Personality (D)** — "the general tendency to maximize one's individual utility — disregarding, accepting, or malevolently provoking disutility for others." [PubMed](#) [Semantic Scholar](#) Bifactor modeling across $N > 2,500$ supports D as a general factor, primarily saturated by Psychopathy (ECV = .74) and Machiavellianism, with Narcissism loading less strongly. [Wildbeimwild](#)

The Dark Tetrad adds **everyday sadism** (Buckels, Trapnell, & Paulhus, 2014). Bonfá-Araujo et al. (2022) meta-analysis: Sadism–Psychopathy $r = .58$, Sadism–Machiavellianism $r = .43$, Sadism–Narcissism $r = .26$. [Uksw](#) Sadism uniquely predicts trolling, cyberbullying, and gratuitous cruelty beyond the Dark Triad. [ScienceDirect](#)

Cross-framework mappings

Big Five (O'Boyle et al., 2015; 310 samples, 215 sources): All three Dark Triad traits correlate negatively with **Agreeableness** [PubMed](#) — this is the universal shared correlate. Narcissism uniquely correlates positively with Extraversion [Wikipedia](#) and negatively with Neuroticism. Psychopathy correlates negatively with both Agreeableness and Conscientiousness. [Wikipedia](#) Machiavellianism and Psychopathy show near-identical Big Five profiles. [Frontiers](#) At the facet level, the FFM explained **88%** of psychopathy variance and **44%** of narcissism variance.

HEXACO: The most striking finding in this literature: latent Dark Triad $\leftrightarrow$ Honesty-Humility at $r = -.95$ (Hodson et al., 2018). [ScienceDirect](#) Observed composite SD3 Dark Triad $\leftrightarrow$ HEXACO-60 HH at $r = -.65$ [95% CI: $-.68, -.62$]. [ScienceDirect](#) The Dark Triad is essentially the inverse of Honesty-Humility. This near-complete overlap means that for LLM research, Honesty-Humility and dark trait dimensions may function as a single bipolar axis.

Linguistic markers and the I-talk debunking

The most important null finding in personality–language research: **first-person singular pronoun use does NOT predict narcissism**. Carey et al. (2015; N=4,811, 5 labs, 5 narcissism measures, 2 languages) found $r = .01$ [95% CI: $-.02, .04$], [Digital Commons](#) definitively overturning Raskin & Shaw's (1988) N=48 finding. This is a critical cautionary tale for any text-based personality detection effort.

Holtzman et al. (2019; 15 samples, N=4,941) found that all LIWC correlates of grandiose narcissism were $|r| < .10$, [Sage Journals](#) and LIWC overall was not more accurate than chance for narcissism prediction (R^2 near zero). A machine learning model achieved only $R^2 = .043$. The strongest narcissism-language signals — sports words $\uparrow$, second-person pronouns $\uparrow$, swear words $\uparrow$, anxiety words $\downarrow$, tentative words $\downarrow$ — are all tiny effects.

[Sage Journals](#)

Psychopathy yields more textually distinctive signals. Hancock et al. (2012) analyzed convicted murderers' speech: more causal conjunctions ("because," "since"), **twice as many** words about physical/material needs (food, sex, money), fewer words about social/spiritual needs, more past tense (psychological distancing), and more speech disfluencies. In online text (Hancock et al., 2018): more hostile language, more anger and swear words, less comprehensible discourse, and more discrepancy-related words.

Machiavellianism shows higher I-words, negative emotion words, and LIWC Clout (confident, high-status writing) in Facebook analysis, along with more analytic language scores. [Taylor & Francis Online](#)

Sadism is the strongest unique predictor of online trolling. Sadism → online aggression at $r = .40$ (pooled). In shared variance models controlling for all Dark Tetrad traits, only psychopathy and sadism predict trolling, with sadism as the strongest predictor.

Key implication for LLM research: Dark Triad traits are extremely difficult to detect from LIWC categories alone. Machine learning provides marginal improvement. The most detectable signals come from psychopathy (hostile, material-focused, emotionally flat language) and sadism (provocative, aggressive content). Narcissism is essentially invisible to dictionary-based text analysis.

3. HEXACO: the sixth factor that captures what Big Five misses

Structure and Big Five correspondence

HEXACO (Ashton & Lee, 2004) adds **Honesty-Humility** (High 5 Test) and rotates the Neuroticism/Agreeableness factor space (Wikipedia) approximately 30°. Thielmann et al. (2022; $k=152$, 6,828 effects) meta-analysis: (Sage Journals) HEXACO Extraversion, Conscientiousness, and Openness map onto Big Five counterparts (ResearchGate) at $r \geq .70$. HEXACO Emotionality ↔ Big Five Neuroticism and HEXACO Agreeableness ↔ Big Five Agreeableness map at $r > .50$ — moderate correspondence reflecting the factor rotation. (Springer) HEXACO Emotionality excludes anger/irritability (which moves to low HEXACO Agreeableness) but includes sentimentality (Springer) (which in Big Five sits under Agreeableness). (Wikipedia +2)

Honesty-Humility has **no direct Big Five counterpart**. (Wikipedia) It correlates with Big Five Agreeableness (Wikipedia) (ScienceDirect) at approximately $r = .26-.67$ (varying dramatically by measure), but captures unique variance in counterproductive work behavior (ScienceDirect) (HEXACO explains **31.97%** of workplace deviance variance vs. Big Five's **19.05%**), (ScienceDirect) (ResearchGate) ethical decision-making, materialism, and — most critically — dark personality traits.

Text-observable signals

HEXACO is **dramatically under-studied** in text analysis. Kleinberg et al. (2020) found HEXACO–LIWC correlations of $r = -.16$ to $.20$, with only 3 of 6 HEXACO traits showing significant linguistic markers. No significant LIWC correlations were found for Honesty-Humility, Agreeableness, or Conscientiousness. (arXiv) Emerging NLP work suggests Honesty-Humility might manifest as straightforward, unembellished phrasing, but this requires validation. **This is a major research gap directly relevant to LLM work.**

4. MBTI: flawed measurement of real constructs

Big Five mappings

McCrae & Costa (1989; $N=468$) established the primary MBTI–Big Five correspondences. (PubMed) MBTI E/I maps onto Big Five Extraversion at $r = .69-.74$ (the strongest mapping). (Interviewaceapp) MBTI S/N maps onto Openness, T/F onto Agreeableness, and J/P onto Conscientiousness (Gustavus Adolphus College) (scirp) — though a large-sample study (Furnham, 2022; $N > 9,000$) failed to replicate the S/N–Openness link, making it the least stable mapping. (scirp) Critically, **no MBTI scale maps onto Neuroticism**, leaving out a major personality dimension. (Wiley Online Library +2)

Validity problems relevant to LLM research

The fundamental issue is forced dichotomization of continuous dimensions: **35% of test-takers** receive a different four-letter type on retest. (Wiley Online Library) (PubMed) Taxometric analyses (Arnau et al., 2003) confirm MBTI dimensions are continuous, not categorical. NLP studies achieving high accuracy on MBTI datasets (up to 86.9%) are likely inflated by dataset artifacts (SpringerOpen) — users self-label in text using MBTI terminology, and reporting accuracy rather than AUC obscures class imbalance problems (PLOS) (Saeteros, Gallardo-Pujol, & Ortiz-Martínez, 2025). (nih)

Practical recommendation for LLM research: Treat MBTI as a proxy for four Big Five dimensions (minus Neuroticism). MBTI-labeled datasets are useful only if treated as noisy continuous variables mapped to Big Five, not as categorical types. Jungian cognitive functions (Ni, Ne, Si, Se, Ti, Te, Fi, Fe) have essentially no empirical validation as distinct measurable constructs.

5. Other well-documented trait systems and their text signatures

Attachment styles

Adult attachment (two-dimensional: Anxiety $\times$ Avoidance) shows highly text-observable patterns. Attachment Anxiety correlates with Neuroticism at $r = .52$ (Ucdavis) (Nofle & Shaver, 2006; meta-analysis K=109, N=32,278) — the strongest cross-framework link. Attachment Avoidance correlates with low Extraversion ($r = -.26$) (Ucdavis) and low Agreeableness ($r = -.22$). (Calstate)

Text markers are strikingly distinct: **Anxious-preoccupied** writers produce more words overall, more negative emotion words, reassurance-seeking phrases ("Are you mad at me?"), rumination markers, and excessive first-person pronouns. (Rolling Out) **Dismissive-avoidant** writers produce fewer words, shorter pragmatic messages, fewer emotional words, more hedging/distancing language ("maybe," "probably," "we'll see"), and emotionally flat prose. (Rolling Out) These opposing profiles — hyperactivation versus deactivation — create one of the most text-detectable trait dimensions available.

Need for Cognition

NFC (Cacioppo & Petty, 1982) correlates with Openness at $r = .40-.50$ and is related to but distinct from Openness-to-Ideas. (Wikipedia) Text markers include: complex sentence structures with subordinate clauses, causal reasoning words ("because," "therefore"), insight words ("think," "know," "consider"), higher LIWC Analytical Thinking scores, and more abstract language. NFC maps directly onto observable linguistic complexity, making it highly tractable for LLM detection.

Regulatory Focus

Higgins' (1997) Promotion versus Prevention focus ([Columbia Business School](#)) produces among the **most directly text-observable** trait signatures. Promotion-focused language uses abstract, interpretive terms, ([ACL Anthology](#)) gain-framing ("achieve," "grow," "aspire"), and approach vocabulary. Prevention-focused language uses concrete, descriptive terms, ([ACL Anthology](#)) loss-framing ("protect," "secure," "avoid"), and vigilance vocabulary. Validated LIWC dictionaries confirm medium-strength correlations between chronic regulatory focus and corresponding word frequencies. ([PubMed Central](#)) Promotion focus correlates with Extraversion ($r > .30$) and Openness ($r > .25$); Prevention focus with Conscientiousness ($r = .20-.30$) and Neuroticism ($r = .20-.30$). ([ScienceDirect](#))

Empathy (Davis IRI)

Empathic Concern maps onto Agreeableness at $\beta = .48$ ([ScienceDirect](#)) (the strongest single association across all empathy–Big Five links). Personal Distress maps onto Neuroticism at $\beta = .53$. ([PubMed Central](#)) Fantasy maps onto Openness at $r = .47$. ([ScienceDirect](#)) ([Springer](#)) Text markers include: perspective-taking language ("I can see how you might feel"), other-oriented emotional language, and — for Personal Distress — self-focused emotional language when discussing others' suffering. Empathic language patterns are among the most readily detectable in text.

BIS/BAS and sensation seeking

Gray's BIS maps onto Neuroticism + low Extraversion ([Academia.edu](#)) ([ScienceDirect](#)) ($r = .40-.55$ with Neuroticism); BAS maps primarily onto Extraversion. ([ScienceDirect](#)) BAS subscales differentiate meaningfully: BAS-Fun Seeking captures impulsive sensation seeking (low Conscientiousness), while BAS-Drive captures goal-directed persistence (high Conscientiousness). ([ScienceDirect](#)) ([Academia.edu](#)) These overlap considerably with Regulatory Focus language patterns — BIS $\approx$ Prevention, BAS $\approx$ Promotion.

Sensation Seeking (Zuckerman) correlates with Extraversion ([Wikipedia](#)) ($r = .35-.50$), Openness ($r = .25-.40$), low Conscientiousness, and low Agreeableness. It is notably **independent of Neuroticism**. Text markers include references to novel, intense, risky experiences, vivid/intense descriptors, and more profanity.

Trait Emotional Intelligence

Trait EI correlates with [ScienceDirect](#) Neuroticism at $r = -.63$ to $-.68$, [Frontiers](#) Extraversion at $r = .35-.45$, and Conscientiousness at $r = .25-.40$. Big Five traits jointly predict **42–59%** of Trait EI variance, [PubMed Central](#) [ScienceDirect](#) raising questions about whether it adds anything beyond the Big Five. For LLM research, Trait EI may be useful as a composite target variable rather than an independent dimension, manifesting as emotional vocabulary richness and cognitive reappraisal language.

Locus of Control

Internal LoC correlates with low Neuroticism ($r = -.20$ to $-.35$), Extraversion ($r = .15-.30$), and Conscientiousness ($r = .20-.35$). Text markers include agentic language with first-person + active verbs ("I decided," "I achieved") for internal LoC, versus passive constructions and references to external forces for external LoC. LIWC-22's Clout dimension partially captures this. Semantic agency analysis in social media text shows low-agency language correlates with depression markers. [ResearchGate](#)

6. The comprehensive cross-framework correlation map

Several trait systems converge on shared dimensions. The following table summarizes the most important cross-framework mappings:

Construct A	Construct B	r (meta-analytic)	Status
Dark Triad (latent)	HEXACO Honesty-Humility	−.95	Very robust
Attachment Anxiety	Big Five Neuroticism	.52	Robust (K=109)
Trait EI (global)	Big Five Neuroticism	−.63 to −.68	Robust
NFC	Big Five Openness	.40–.50	Robust
MBTI E/I	Big Five Extraversion	.69–.74	Robust
BIS	Big Five Neuroticism	.40–.55	Robust
BAS	Big Five Extraversion	.30–.45	Robust
Empathic Concern	Big Five Agreeableness	$\beta = .48$	Robust (cross-cultural)
Personal Distress	Big Five Neuroticism	$\beta = .53$	Robust
Fantasy (IRI)	Big Five Openness	.47	Robust
Sensation Seeking	Big Five Extraversion	.35–.50	Robust
All Dark Triad traits	Big Five Agreeableness	−.25 to −.47	Very robust
Psychopathy	Big Five Conscientiousness	negative	Robust
Narcissism	Big Five Extraversion	positive	Robust
Mach–Psychopathy overlap	—	$\rho = .52$ (80% shared variance)	Robust but contested

This convergence map reveals a critical structural insight: **much of the cross-framework correlation structure is explained by the Big Two metatraits**. Stability (high A, high C, low N) captures the prosocial/self-regulated pole (ScienceDirect Umn) — its inverse maps onto Dark Triad, external LoC, insecure attachment, low Trait EI, and high Personal Distress. Plasticity (high E, high O) captures the exploratory/engaged pole (ScienceDirect Umn) — mapping onto high NFC, Promotion focus, Sensation Seeking, high BAS, and Fantasy.

7. Text analysis methodology: what works and what doesn't

The accuracy hierarchy

Personality prediction from text follows a clear accuracy gradient. Individual LIWC categories correlate with self-reported Big Five at $r = .05-.14$ (explaining $< 2\%$ variance per category). All 52 LIWC categories combined explain roughly **5.1%** of self-reported personality variance. (ResearchGate +2) Open-vocabulary machine learning reaches $r = .31-.41$ (ResearchGate) (Schwartz et al., 2013; Park et al., 2015). (UC Irvine Language Science) LLM zero-shot prediction achieves $r = .22-.33$ (Peters & Matz, 2024, PNAS Nexus; GPT-4), (arXiv) matching purpose-built supervised models. The practical ceiling for text $\rightarrow$ self-reported Big Five appears to be $r \approx .35-.45$.

Crucially, observer-reported personality is far more predictable from text ($\rho = .18-.39$ for individual LIWC categories; 38.5% total variance) than self-reported personality (5.1% variance). (Vrije Universiteit Amsterd...) This suggests text reflects personality *as perceived by others* more than internal self-concept — a distinction directly relevant to LLM personality simulation.

The seven most robust text–personality associations

These associations replicate across Pennebaker & King (1999), Mehl et al. (2006), Hirsh & Peterson (2009), Yarkoni (2010), Schwartz et al. (2013), and Koutsoumpis et al. (2022):

1. **Neuroticism** ↔ **negative emotion words** — the single most replicated finding
2. **Neuroticism** ↔ **first-person singular pronouns** — well-replicated
3. **Extraversion** ↔ **positive emotion words** — robust across settings
4. **Extraversion** ↔ **social process words** — consistent
5. **Openness** ↔ **word length and vocabulary complexity** — robust
6. **Openness** ↔ **tentative/hedging words** — consistent
7. **Low Agreeableness** ↔ **swear words** — robust

What fails to replicate or is weakly supported

Conscientiousness shows the weakest language markers and is the hardest Big Five trait to predict from text. **Narcissism** ↔ **I-talk** is definitively debunked ($r = .01$; Carey et al., 2015). [Digital Commons](#) Past tense usage shows inconsistent associations. Punctuation patterns are under-studied and unreliable. Cross-domain generalizability is a major concern — models trained on essays degrade when applied to Reddit (Berggren et al., 2024), and LIWC–personality associations found in training data fail to generalize to holdout samples in some studies [MDPI](#) (Martínez-Huertas et al., 2022). [MDPI](#) [Diva-portal](#)

Critical methodological constraints for LLM finetuning

Text volume requirement: Several hundred to thousands of words per person are needed for reliable personality signal. [arXiv](#) Single short texts capture personality *state*, not stable *trait*. Peters & Matz (2024) showed prediction accuracy increases monotonically with message volume.

Genre effects: Personality manifests differently across blogs (richest signal, voluntary, topic-diverse), [nih](#) social media (shorter, modulated by platform norms), essays (constrained context), and formal writing (personality signal attenuated by genre conventions). **Trait activation theory** explains this: traits show up in language primarily when situational cues are relevant — Neuroticism markers emerge during stress, Extraversion markers during social contexts.

The most promising non-Big Five dimensions for LLM detection are Regulatory Focus (clean promotion/prevention linguistic signatures), Attachment Style (strikingly distinct anxious vs. avoidant communication profiles), Need for Cognition (direct mapping to syntactic complexity), and Empathy subscales (perspective-taking and compassion language readily detectable). These may outperform Big Five dimensions in text observability because they produce more behaviorally specific language patterns rather than diffuse statistical tendencies.

Conclusion: what this means for LLM finetuning design

Three structural insights should guide finetuning experiments. First, **personality dimensions are not independent** — they cluster into the Big Two (Stability and Plasticity), and many trait systems outside the Big Five map onto these same two axes. (ScienceDirect) (Sage Journals) Finetuning on a single trait like Extraversion will likely shift Openness in the same direction and Neuroticism in the opposite direction. Cross-trait correlations should be monitored, not just target traits. Dark Triad traits are essentially the inverse of Honesty-Humility, (Liberty University +2) which is itself partially captured by Big Five Agreeableness — so manipulating Agreeableness in an LLM will likely shift dark personality expression.

Second, **open-vocabulary approaches dramatically outperform dictionary-based methods** (Academia.edu) ($r = .31-.41$ vs. $.21-.29$), (UC Irvine Language Science) and LLMs already match purpose-built supervised classifiers in zero-shot personality assessment. (Emergent Mind +2) This suggests LLMs already encode substantial personality-language knowledge and that finetuning should focus on reinforcing the most robust associations (emotion words, pronouns, complexity, social words) while avoiding training on weak or debunked signals (I-talk for narcissism, past tense, specific punctuation).

Third, **observer-rated personality is far more text-accessible than self-reported personality** (38.5% vs. 5.1% variance explained by LIWC). (Vrije Universiteit Amsterd...) For LLM applications where the goal is to generate text that *reads as* having a particular personality, observer-rated personality profiles are a more appropriate training target than self-report scores. The practical ceiling of $r \approx .35-.45$ for self-report prediction reflects a genuine measurement boundary — text captures personality as socially expressed, not as privately experienced. This distinction is a feature, not a bug, for LLM personality simulation.